

OROVILLE, CA, USA

WMO: 745048

Lat: 39.490N

Lon: 121.618W

Elev: 190

StdP: 14.60

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 93210

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	33.1	35.6	18.1	13.8	51.6	22.1	16.7	50.1	27.7	51.5	24.9	51.0	3.0	320	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.3	100.5	69.8	98.1	69.2	95.2	67.9	73.3	95.0	71.5	93.3	70.1	91.5	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	94.9	86.2	63.2	87.3	84.5	61.3	81.4	82.2	37.0	95.3	35.4	93.4	34.0	91.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	18.2	16.1	DB	28.1	106.4	2.5	2.4	26.3	108.1	24.8	109.6	23.4	110.9	21.6	112.7
			WB	26.3	77.2	2.2	4.7	24.8	80.6	23.5	83.4	22.3	86.1	20.7	89.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	47.7	51.2	55.7	60.2	68.0	75.7	80.5	78.3	73.7	64.6	54.0	47.7	(6)
	DBStd	12.82	4.58	4.76	5.46	6.55	6.77	6.63	5.22	4.43	5.57	5.59	5.70	4.96	(7)
	HDD50	282	97	39	12	3	0	0	0	0	0	0	27	104	(8)
	HDD65	2388	535	387	292	174	44	6	0	0	5	75	332	538	(9)
	CDD50	5091	27	72	188	308	558	772	946	878	711	452	147	32	(10)
	CDD65	1720	0	0	3	29	137	328	481	413	266	62	1	0	(11)
	CDH74	22120	1	1	78	459	1772	4242	6453	5110	3184	799	19	2	(12)
	CDH80	11361	0	0	7	136	728	2245	3744	2738	1526	236	1	0	(13)
(14) Wind	WSAvg	6.1	5.3	6.7	7.1	7.1	7.1	6.7	6.0	5.6	5.4	5.2	5.0	5.8	(14)
(15) Precipitation	PrecAvg	20.9	4.3	4.0	3.0	1.6	1.3	0.4	0.0	0.1	0.1	1.1	1.9	4.5	(15)
	PrecMax	35.5	14.1	11.5	10.4	4.9	4.0	1.5	0.1	0.6	0.6	3.7	5.0	11.1	(16)
	PrecMin	12.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	(17)
	PrecStd	5.9	3.4	2.8	3.2	1.4	1.1	0.4	0.0	0.2	0.2	0.9	1.4	2.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	72.1	81.0	89.2	96.8	103.5	105.8	102.1	99.3	91.1	76.6	66.2	(19)
		MCWB	50.6	56.7	60.4	63.4	69.3	71.1	71.5	69.2	67.6	64.1	58.5	53.7	(20)
	2%	DB	61.8	67.4	75.8	83.9	91.9	99.1	101.5	98.7	95.3	86.0	72.2	61.3	(21)
		MCWB	50.7	55.2	58.5	61.5	66.7	69.8	70.4	68.7	66.1	61.5	57.3	53.0	(22)
	5%	DB	58.5	63.6	72.2	79.8	88.2	95.3	98.8	95.5	92.5	81.9	68.0	58.6	(23)
		MCWB	51.1	53.2	57.1	60.3	64.8	67.7	69.9	67.7	65.6	60.1	55.6	52.2	(24)
	10%	DB	55.6	60.6	67.9	74.6	83.6	91.4	95.5	92.7	89.3	78.7	63.9	56.2	(25)
		MCWB	50.2	52.1	54.9	58.1	62.8	66.4	68.8	67.3	65.0	58.9	54.4	50.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	59.8	62.5	66.5	75.3	75.6	76.1	74.4	72.8	67.0	61.8	58.9	(27)
		MCDB	60.7	67.9	76.2	83.0	88.1	96.4	99.6	92.9	89.7	84.3	72.5	63.3	(28)
	2%	WB	54.2	57.2	60.3	63.5	69.4	72.2	73.4	71.5	69.1	64.3	59.3	56.4	(29)
		MCDB	57.7	64.3	72.9	80.8	87.6	93.8	96.6	92.8	90.2	80.8	67.6	59.6	(30)
	5%	WB	52.7	55.2	58.4	61.2	66.4	70.2	71.8	70.0	67.3	62.4	57.7	54.0	(31)
		MCDB	55.9	61.1	69.1	77.0	85.0	91.4	94.6	91.1	88.4	77.3	64.1	56.3	(32)
	10%	WB	51.1	53.4	56.5	59.2	64.0	68.2	70.4	68.5	65.7	60.5	56.1	52.1	(33)
		MCDB	55.1	58.6	65.7	72.7	81.4	88.8	92.6	89.5	86.9	74.0	62.2	54.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	17.0	19.6	22.4	24.9	27.3	29.3	28.8	27.6	24.4	17.9	14.7	(35)
		MCDBR	21.0	23.4	27.5	30.6	30.7	31.3	31.2	30.7	32.0	30.5	24.7	18.5	(36)
	5% WB	MCWBR	12.6	13.7	14.0	13.3	12.4	11.3	11.4	11.3	12.4	12.8	12.6	11.1	(37)
		MCWBR	14.1	18.6	23.5	28.4	28.7	29.7	29.7	29.0	29.5	26.1	19.0	11.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.328	0.329	0.351	0.360	0.360	0.356	0.353	0.340	0.343	0.339	0.323	0.327	(40)
		taud	2.485	2.494	2.426	2.397	2.435	2.448	2.439	2.470	2.473	2.486	2.517	2.489	(41)
	Ebn at Noon		267	284	287	290	291	291	291	293	286	276	266	258	(42)
		Edh at Noon	25	28	33	36	36	36	36	34	31	28	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		671	968	1380	1863	2301	2605	2623	2335	1872	1288	800	581	(44)
	RadStd		121	137	170	185	157	119	82	79	72	101	72	104	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (32 neighbors)	+0.74	N/A	-2.00	N/A	N/A	N/A	-58	-180	+241	+90

Nomenclature: See separate page